# Brain Microstructural Alterations in Primary Headache and Their Association with Mental Health Status

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# Problem Statement

- Brain mechanisms underlying headache remain unclear
- Headache affects more than pain alone
- Its impact can extend to mental and emotional well-being
- Therefore, understanding headache requires a broader perspective beyond physical symptoms

# Goal



Explore whether mental health is associated with headache



Assess how regional DTI metrics are associated with mental health

# Diffusion Tensor imaging

- DTI (Diffusion Tensor Imaging) is a type of MRI technique
- Measures how water molecules move in brain tissue
- Water moves differently in organized fiber pathways
- Helps evaluate brain microstructure and connectivity

## Rationale for Using DTI

- DTI can help identify whether pathways involved in emotional regulation show microstructural alterations (1)
- Low anisotropy and the lack of a strong principal fiber direction make FA less informative and AD and RD less reliable in gray matter
- MD is more suitable for capturing subtle gray-matter microstructural variation

1. Aggarwal, *Translational Psychiatry* 12.1 (2022):

# Mental Component Summary (MCS)

- It reflects overall mental and emotional health status
- Assesses how emotional health affects daily activities
- Helps estimate overall mental health–related quality of life

# Literature Review

**1.**
SF-36 MCS measures psychological symptom burden and reflects mental health status. (5)

**2.**
DTI detects microstructural brain changes non-invasively (1,2,3)

**3.**
Microstructural changes in specific brain pathways are associated with mental-health status, and DTI abnormalities may reflect brain changes related to emotional well-being (4)


1. Yang HC. Diagnostics (Basel). 2025 Mar 6;15(5):
2. Ellingson BM. Pain. 2013 Sep;154(9):1528-1541. doi: 10.10
3. Nanda C. Neurol India. 2024 Nov 1;72(6):1169-1173. doi: 10.4103/ni.ni_894_22. Epub 2024 Dec 17. PMID: 39690987.
4. Rahimi R. Brain Imaging Behav. 2022 Oc
5. Pfoh ER. Ann Am Thorac Soc. 2016 Aug;13(8):1343-50. doi:

# Bridging the Research Gap

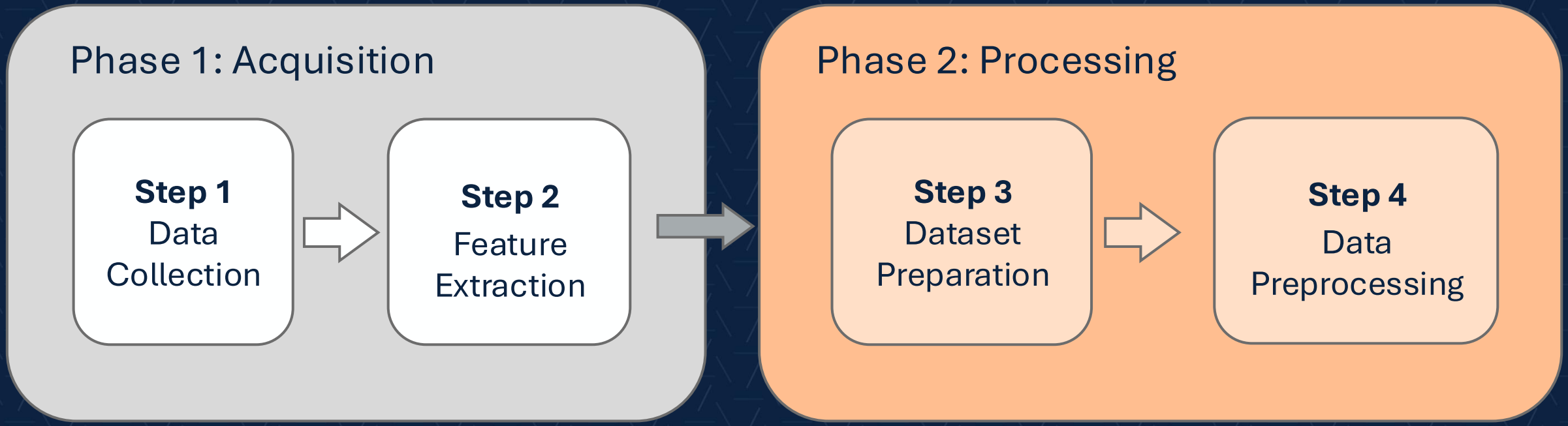
| Previous studies | Limitation | Current study |
| --- | --- | --- |
| Focused on pain outcomes in brain with headache | Emotional burden was not well explored | Investigates the relationship between mental/emotional status and headache |
| Reported mainly global imaging findings | Specific regional associations with mental-health status were unclear | Examines region-specific DTI correlates of MCS |

# Workflow

Part 1:
DATA Preparation

Part 2:
Data Correlation

# Methodology



# Phase 1: Acquisition

**Cohort Recruitment**

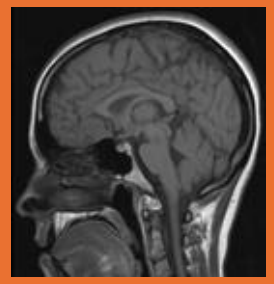


15 Primary Headache Patients

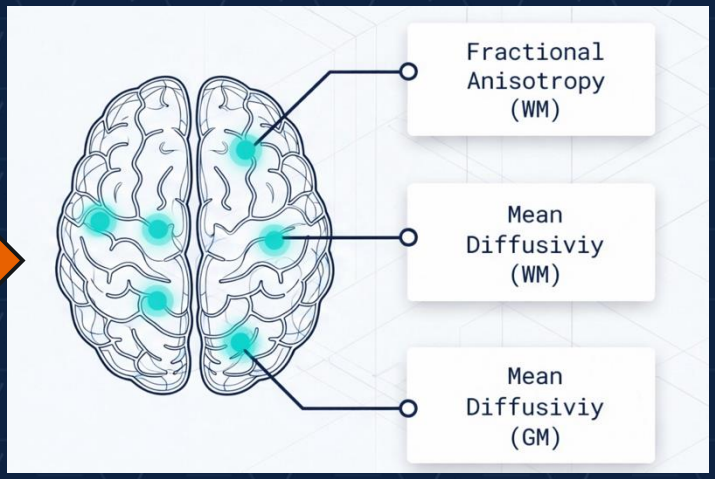
**Baseline Scan**



Siemens 7T Ultra High Field MRI

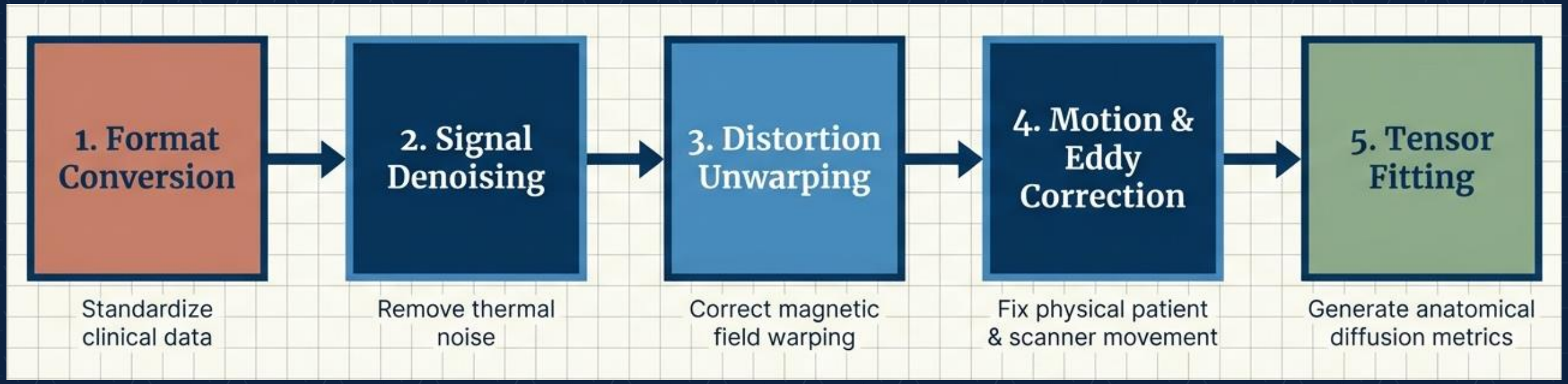


Data of PH Patients



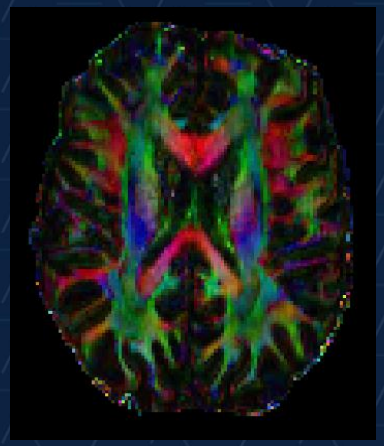

DTI Features

# Phase 1: Acquisition (DTI Features)

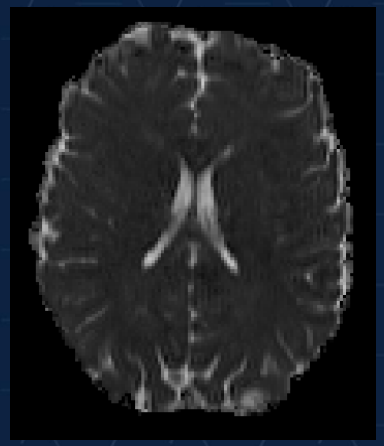



**Software and Processing Tools:** FSL and NORDIC

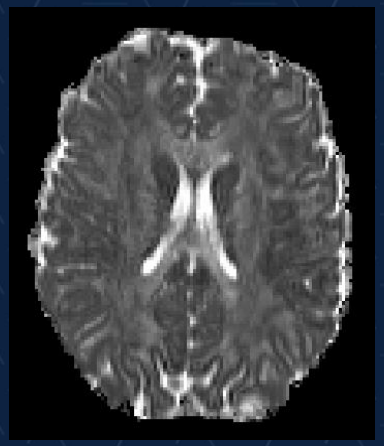
# Phase 1: Acquisition (DTI Features)



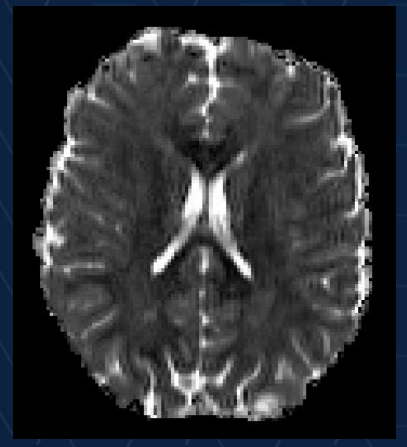
Fractional Anisotropy (FA)



Mean Diffusivity (MD)

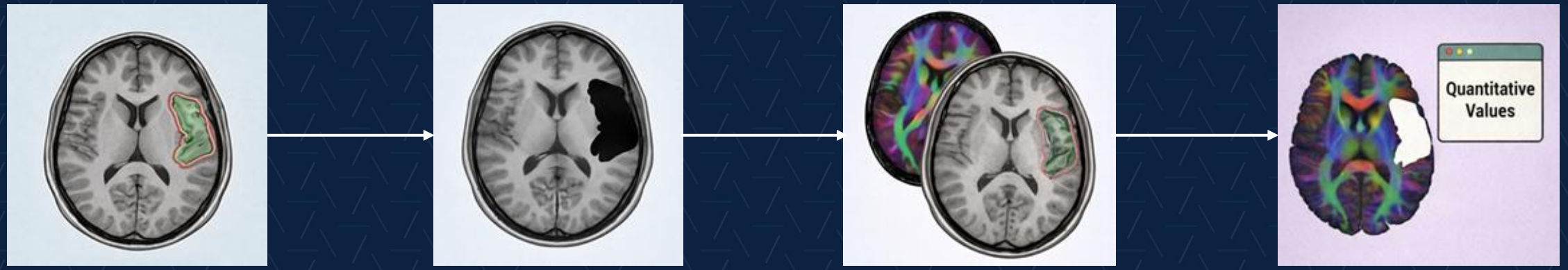


Axial Diffusivity (AD)



Radial Diffusivity (RD)

# Phase 2: Processing




Define ROI on T1 Structural Image

Binary ROI Mask

Align Maps and T1

Mean Metric Values

**Software and Processing Tools:** Freesurfer

# Phase 3: Statistical Analysis

- Spearman correlation evaluates the strength and direction of association between two variables using ranked values
- Performed between each DTI feature of patients and corresponding MCS
- FDR correction was applied to account for multiple comparisons
- Both uncorrected and FDR-corrected significant associations were reported

# White Matter Findings

| No | Hemisphere | Region | Spearman rho | p_value | DTI metric | MCS |
|---|---|---|---|---|---|---|
| 1. | L | Temporal | -0.69 | 0.0038 | ↑ **RD** | ↓ |
| ** NO Region survived after FDR Correction | | | | | | |
| 4. | R | Occipital | 0.57 | 0.0238 | ↑ **AD** | ↑ |
| 5. | L | Occipital | -0.56 | 0.0272 | ↑ **RD** | ↓ |
| 6. | L | Parietal | -0.53 | 0.0409 | ↑ **MD** | ↓ |

# White Matter Findings

- Higher RD may reflect reduced white matter integrity associated with poorer mental health
- Higher AD with higher MCS may suggest better microstructural integrity in visual cortex related to better mental health
- Higher parietal MD with lower MCS may indicate reduced microstructural integrity (left parietal region)

# Gray Matter Findings

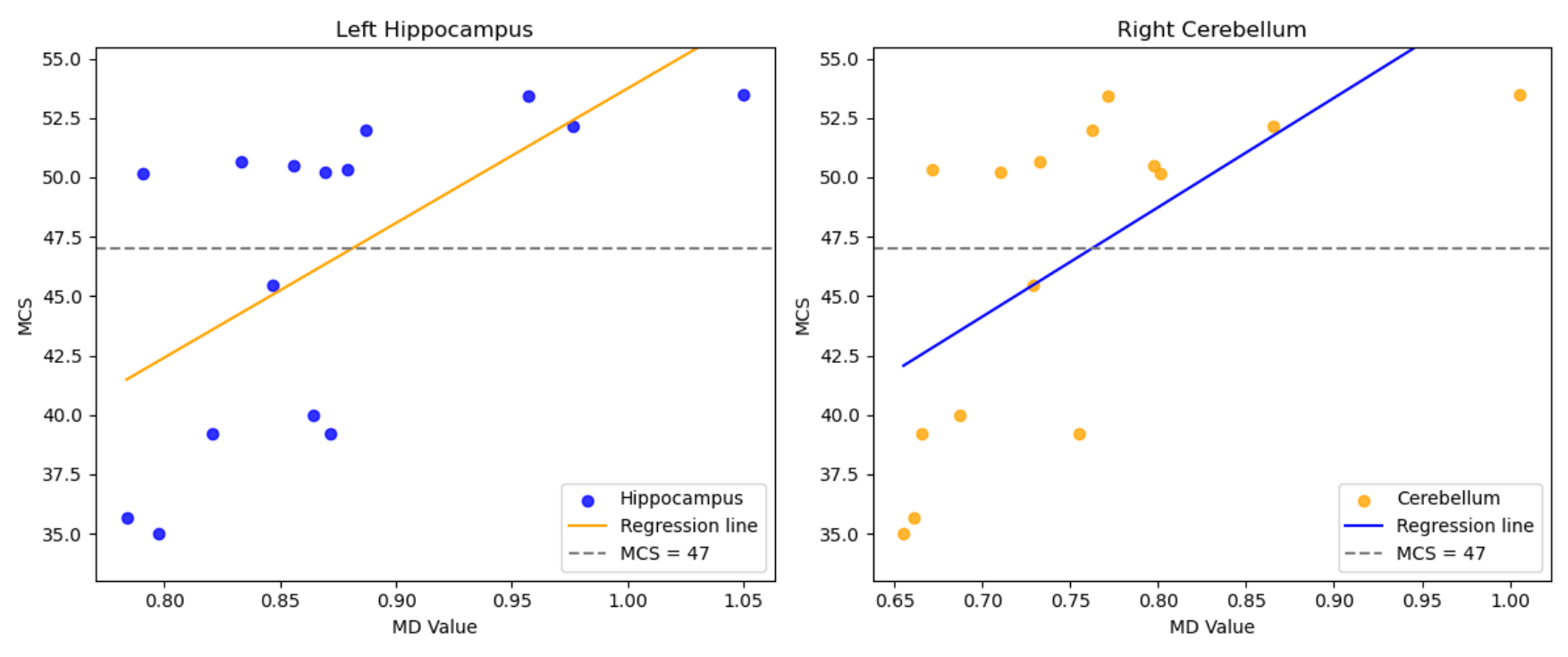
| No | Hemisphere | Region | Spearman rho | p_value | DTI metric | MCS |
|---|---|---|---|---|---|---|
| 1. | R | Cerebellum | 0.78 | 0.0004 | ↑ MD | ↑ |
| 2. | L | Hippocampus | 0.77 | 0.0007 | ↑ MD | ↑ |
| *** FDR corrected significant Region | | | | | | |
| 4. | R | Occipital | 0.57 | 0.0238 | ↑ **MD** | ↑ |
| 5. | L | Pallidum | -0.53 | 0.0395 | ↑ **MD** | ↓ |

# Gray Matter Findings

- Higher MD in the cerebellum, hippocampus, and occipital gray matter was associated with higher MCS, suggesting better mental well-being
- Higher MD in the left pallidum was associated with lower MCS, suggesting poorer emotional well-being

# Gray Matter Findings

- Increased cerebellar MD might positively associated with improved cognitive performance in certain cases (1)




1. Aggarwal, *Translational Psychiatry* 12.1 (2022):

# Limitations

- This is an exploratory study with a small sample size
- The dataset is heterogeneous, including two primary headache subtypes: tension-type headache and migraine
- The relationship between mental health and headache subtype may be non-linear and requires further evaluation

# Future Contributions

- Assess whether MCS-related brain regions are similar across group
- Investigate whether headache patients have higher emotional burden than controls

# Conclusion

- increases in MD may reflect beneficial microstructural remodeling, which may contribute to improved cognitive performance.
- Cerebellar gray matter contributes to cognition, attention, and emotional regulation, and its microstructural alterations may be associated with mental health status
- The hippocampus is involved in memory, emotion, and stress-related processing and it may have positive association with mental well-being